

```
Installing package into '/home/shakthi/R/x86_64-pc-linux-
gnu-library/4.1'
(as 'lib' is unspecified)
also installing the dependencies 'gert', 'usethis'
trying URL 'https://cloud.r-project.org/src/contrib/
gert_1.5.0.tar.gz'
Content type 'application/x-gzip' length 66958 bytes (65 KB)
...
** testing if installed package keeps a record of temporary
installation path
* DONE (devtools)
> library(devtools)
Loading required package: usethis
```

You can now use the `install_github` function with a GitHub 'username/project-name' to install the R sources from the main branch, as shown below:

```
> install_github("hadley/plyr")
Downloading GitHub repo hadley/plyr@HEAD
...
- preparing 'plyr':
✓ checking DESCRIPTION meta-information ...
- cleaning src
- checking for LF line-endings in source and make files and
shell scripts
- checking for empty or unneeded directories
- building 'plyr_1.8.6.9000.tar.gz'
** checking absolute paths in shared objects and dynamic
libraries
** testing if installed package can be loaded from final
location
** testing if installed package keeps a record of temporary
installation path
* DONE (plyr)
```

R also provides the `package.skeleton` function to build your own package from a directory containing R source files. This function accepts the following arguments:

Argument	Purpose
name	Package name
list	A character vector of named R objects
environ- ment	The environment (default .GlobalEnv) to evaluate list
path	A character vector (default '.') containing path in the file system
force	A Boolean value to overwrite files if directory already exists
namespace	A Boolean value to include a namespace
code_files	The character vector that mentions the R source files

Repositories

The R packages are available on the Internet in three major repositories:

1. **Comprehensive R Archive Network (CRAN):** This provides source code as well as pre-compiled binary distributions for the R base system and packages. The latest R release is 2021-11-01 (Bird Hippie) R-4.1.2.tar.gz as of writing this article. It has over 18,928 packages, and you can view the available packages by name or date of publication.
2. **Bioconductor:** This project provides Free/Libre and Open Source R software packages for powerful statistical and graphical methods to analyse genomic data. These packages are also made available in CRAN, in addition to their own R packages. Their software is released under the Artistic License 2.0.
3. **R-Forge:** This provides a platform for the development of R packages and hosts 2145 R related projects. It is based on FusionForge, and uses Subversion for source control management. It has support for mailing lists, forums, bug tracking, Web based administration and backups.

The remote repositories available on your system can be obtained using the following `options` function:

```
> options('repos')
$repos
CRAN
"https://cloud.r-project.org"
```

You can add additional remote R repositories using the `setRepositories` function. 

References

- [1] The Comprehensive R Archive Network (CRAN), <https://cran.r-project.org/>
- [2] Bioconductor, <https://www.bioconductor.org/>
- [3] R-Forge, <https://r-forge.r-project.org/>
- [4] FusionForge, <https://fusionforge.org/>

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